Mohammed Shafiullah Raihan

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shafiullah-raihan | Shafiullah-Raihan

Profile

Computer Science and Engineering graduate with strong foundations in data structures, algorithms, and software development, and **3,600+ programming problems solved** across online judges. Author of **two research papers**: a co-authored preprint on Transformer-based image captioning and a PLS-SEM study of AI adoption among 460 healthcare professionals in Bangladesh. Built projects in C++, Python, Django, JavaScript, SQL, and networking, and completed a Machine Learning thesis on network intrusion detection. Seeking a **Software Engineer** role where I can apply algorithmic thinking, clean implementation, and ML/data skills to real-world problems.

Education

International Islamic University, Chittagong *April 2019 – January 2026*
BSc in Computer Science and Engineering

- **CGPA:** 2.994 / 4.00
- **Relevant Coursework:** Data Structures & Algorithms, Computer Programming, Computational Theory, Database Systems, Operating Systems, Computer Networks, Software Engineering, Machine Learning

Research Publications

A CNN-Transformer Encoder-Decoder for Image Captioning: Label Smoothing, Backbone Fine-Tuning, and Length-Normalized Decoding *October 2026* *Preprint (DOI)*

- R. H. Shawon, F. A. F. Siam, **Mohammed Shafiullah Raihan**, S. Khan, M. S. Munna, M. M. Azad, M. Arman, S. Kim, H. Byeon. *Research Square* (preprint, not peer reviewed).
- InceptionV3 + Transformer captioner trained on MS COCO; ablations showed beam-search decoding to be the most influential component (CIDEr 0.7495 for the full system), and documented a BatchNorm-freezing pitfall in partial CNN fine-tuning.

Artificial Intelligence Readiness and Technology Adoption among Healthcare Professionals in Bangladesh: Evidence from PLS-SEM *2026 (forthcoming)*

- **Mohammed Shafiullah Raihan** (sole author). *E-Palli Publishers* (accepted).
- Survey of **460 healthcare professionals**; seven constructs and 28 indicators modelled with PLS-SEM. The model explains 45% of variance in AI readiness and 43% in reported AI adoption.

Technical Projects

Network Intrusion Detection System (NIDS) — BSc Thesis *GitHub*

- Developed a machine-learning-based NIDS covering data preprocessing, exploratory analysis, feature engineering, supervised classification, and model evaluation.
- **Tools:** Python, Jupyter Notebook, Pandas, NumPy, Scikit-learn, Matplotlib, Git/GitHub.

Hybrid Multi-Level Queue CPU Scheduling Simulator *GitHub — Live Link*

- Built an interactive simulator for FCFS, SJF, Priority, Round Robin, and Hybrid Multi-Level Queue scheduling with Gantt-chart visualization and scheduling metrics.
- **Tools:** HTML5, CSS3, Vanilla JavaScript.

Django Web Applications *Store | Website*

- Developed two full-stack Django applications (an e-commerce platform and a dynamic responsive website) with database integration, CRUD functionality, and responsive UI.
- **Tools:** Python, Django, HTML, CSS, Bootstrap, JavaScript, SQL.

BGP Hijack Attack Simulation & Mitigation *GitHub*

- Simulated BGP route hijacking, malicious route injection, traffic redirection, and recovery in a multi-AS network, and analyzed the security implications.
- **Tools:** Cisco IOS, BGP, Cisco Packet Tracer, VPCS, Cisco CLI.

Industrial Automation & Smart Monitoring *GitHub*

- Built an IoT-based industrial monitoring and automation prototype using sensors, controllers, and data monitoring. **Tools:** C/C++, Python, Arduino/ESP, embedded systems.

Competitive Programming & Problem Solving

- Solved **3,600+ programming problems** across Codeforces, LeetCode, CodeChef, GeeksForGeeks, VJudge, CSES, AtCoder, SPOJ, LightOJ, Beecrowd, and other judges.
- **Problem-solving areas:** Data Structures, Graph Algorithms, Dynamic Programming, Greedy, Binary Search, Shortest Paths, BFS/DFS, Trees, DSU, Number Theory, Prefix Sums, Sliding Window, Two Pointers, Recursion, Backtracking, and Complexity Optimization.
- Strong experience in turning algorithmic ideas into efficient **C++** implementations and debugging time/memory-limit issues.

Codeforces	Shafiullah_Raihan	**LeetCode**	shafiullah-raihan
CodeChef	ms_raihan	**AtCoder**	ms_raihan
CSES	s-raihan	**VJudge**	ms_raihan
GeeksForGeeks	shafiullahraihan	**More**	SPOJ | LightOJ | Beecrowd | UVa

Certifications

Web Development With Python, Django *GitHub*
The Universe IT

Full Stack Web Development with Python and JavaScript *GitHub*
Bohubrihi

Technical Skills

Programming: C, C++, Python, JavaScript
Web Development: HTML, CSS, Bootstrap, JavaScript, React, Django, REST API
Database: SQL, MySQL, Oracle
Algorithms: Data Structures, Graphs, Dynamic Programming, Greedy, Binary Search, Trees, DSU, Shortest Path, BFS/DFS, Number Theory
Machine Learning & Research: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook, Data Preprocessing, Classification, Model Evaluation, PLS-SEM, Survey Data Analysis
Tools: Git, GitHub, Cisco Packet Tracer, GNS3, VS Code, Jupyter Notebook